Escaping Spectral Packets and the Missing Endpoint-Energy Tightness

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Abstract

We construct genuine finite spectra whose moments converge exactly at every fixed order and whose logarithms converge on every strict annulus, while the endpoint H^2 norm stays bounded below. The construction adds a single escaping root-of-unity packet to the exact prefix spectra. Its mass fits the archived inverse-noise cap after reparametrization. This proves that the coarse modulus, mass, fixed-order, and interior-annulus axioms do not decide endpoint convergence. It does not prove that the actual noisy family follows the counterexample.

1 Escaping packet construction

Let $\mathcal{S}_{K-1}$ be an RH-316 spectrum matching the deterministic moments through order $K-1$, chosen with the controlled multiplicities of RH-317. Its endpoint mismatch coefficient at order K is real. Choose a sign aligned with that coefficient and add the RH-315 K-packet with selected moment

$$w_K = \pm K q_*^K.$$

Put $R = 7/5$, $s = q_*/q$, and take $L_K = \lceil s^K \rceil$. This places the packet in $|\mu| \le q$. It changes no moment below K, while its endpoint logarithmic coefficient at order K is exactly ± 1.

Theorem 1.1 (interior convergence with endpoint failure)**.** *There is a sequence of finite conjugate spectra and noise labels σ_K such that:*

1. *the squared mass is at most σ_K^{-1};*
2. *every fixed power sum eventually equals the deterministic anchor;*
3. *the logarithmic mismatch tends to zero in $H^\infty(\rho)$ and $H^2(\rho)$ for every $\rho < \rho_*$;*
4. *the endpoint $H^2(\rho_*)$ norm is at least one.*

Moreover $\log(1/\sigma_K) = K \log(q_*/q) + O(\log K)$.

Proof. The prefix property and packet invisibility give the fixed moments. The packet radius and squared mass are

$$r_K = q_* L_K^{-1/K} \le q, \qquad B_K = K L_K r_K^2 = K q_*^2 L_K^{1-2/K} = \Theta(K s^K).$$

Combine it with the $O(s^K)$ prefix mass and define σ_K as the reciprocal of one plus the total mass. This proves both the mass cap and the stated logarithmic clock. On a strict radius, the first packet coefficient is $(q_*\rho)^K$ and the coefficient at order mK is

$$\frac{(q_*\rho)^{mK}}{m L_K^{m-1}}.$$

Their H^∞ coefficient sum and H^2 energy both vanish. The prefix mismatch vanishes by RH-319 when $R < \rho < \rho_*$. For a smaller radius, apply that result first at any intermediate ρ' with $\max\{R, \rho\} < \rho' < \rho_*$ and then restrict the norm. At the endpoint, choose the packet sign to align with the pre-existing order-K mismatch; that single coefficient then has modulus at least one. □

2 Exact tightness criterion

Let

$$e_{\sigma,n} = (\tau_{\sigma,n} - a_n)\rho_*^n/n.$$

Corollary 2.1 (endpoint energy tightness)**.** *The endpoint mismatch tends to zero in H^2 if and only if*

$$e_{\sigma,n} \to 0 \quad \textit{for every fixed } n$$

and

$$\lim_{N\to\infty} \limsup_{\sigma\downarrow 0} \sum_{n>N} |e_{\sigma,n}|^2 = 0.$$

Proof. This is the standard finite-head plus uniformly tight-tail characterization of norm convergence in ℓ^2 under the Taylor-coefficient isometry. □

Remark 2.2. The repository has neither actual fixed-order complement transport nor the tail-tightness estimate. The synthetic counterexample proves logical insufficiency only, not actual noisy nonconvergence. Gates A–E remain false/open.